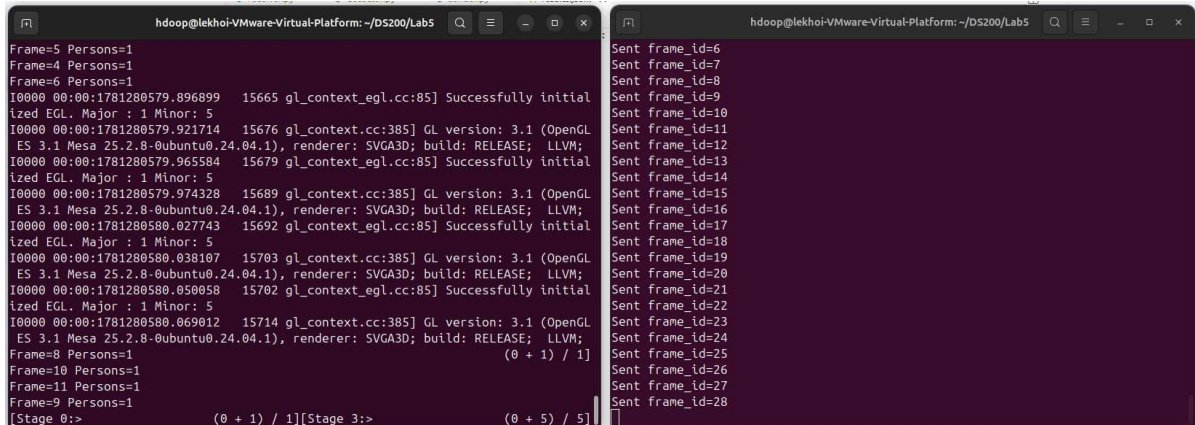


Do em xài máy ảo nên việc thiết lập camera để detect trở nên lằng nhằng nên em thay thế đầu vào bằng video. Tức là sender sẽ nhận video và gửi các frame đã được encode cho receiver xử lý

Em có thêm tác vụ vẽ bounding box và tại mỗi thời điểm, thay vì trả về output chỉ là json thì em trả về khung hình đó luôn.

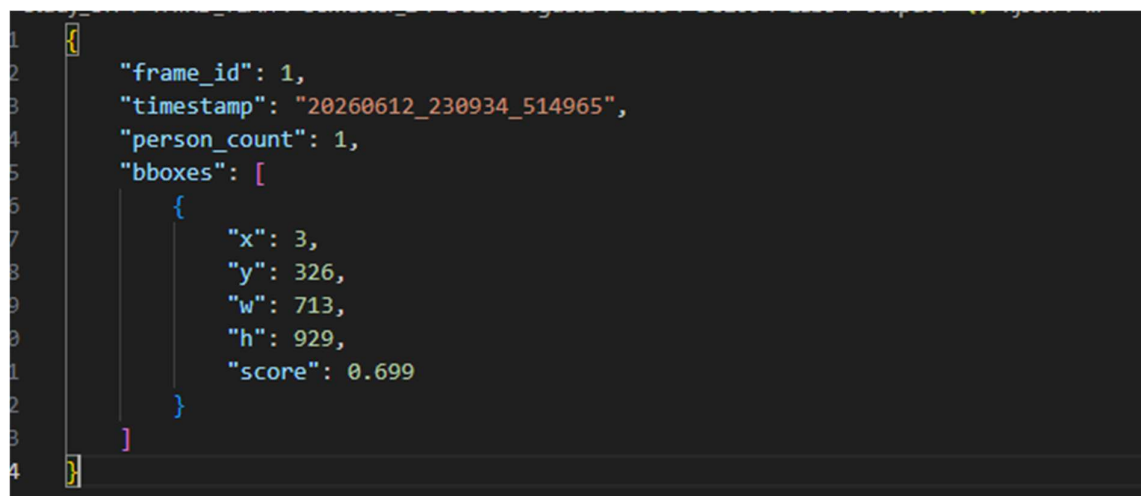


The image shows two terminal windows from a VMware virtual platform. The left window displays system logs for OpenGL context initialization and frame processing. The right window shows a list of frame IDs being sent, ranging from 6 to 28.

```
hdoop@lekhoi-VMware-Virtual-Platform: ~/D5200/Lab5
Frame=5 Persons=1
Frame=4 Persons=1
Frame=6 Persons=1
10000 00:00:1781280579.896899 15665 gl_context_egl.cc:85] Successfully initial
ized EGL_Major : 1 Minor: 5
10000 00:00:1781280579.921714 15676 gl_context.cc:385] GL version: 3.1 (OpenGL
ES 3.1 Mesa 25.2.8-0ubuntu0.24.04.1), renderer: SVGA3D; build: RELEASE; LLVM;
10000 00:00:1781280579.965584 15679 gl_context_egl.cc:85] Successfully initial
ized EGL_Major : 1 Minor: 5
10000 00:00:1781280579.974328 15689 gl_context.cc:385] GL version: 3.1 (OpenGL
ES 3.1 Mesa 25.2.8-0ubuntu0.24.04.1), renderer: SVGA3D; build: RELEASE; LLVM;
10000 00:00:1781280580.027743 15692 gl_context_egl.cc:85] Successfully initial
ized EGL_Major : 1 Minor: 5
10000 00:00:1781280580.038107 15703 gl_context.cc:385] GL version: 3.1 (OpenGL
ES 3.1 Mesa 25.2.8-0ubuntu0.24.04.1), renderer: SVGA3D; build: RELEASE; LLVM;
10000 00:00:1781280580.050858 15702 gl_context_egl.cc:85] Successfully initial
ized EGL_Major : 1 Minor: 5
10000 00:00:1781280580.069812 15714 gl_context.cc:385] GL version: 3.1 (OpenGL
ES 3.1 Mesa 25.2.8-0ubuntu0.24.04.1), renderer: SVGA3D; build: RELEASE; LLVM;
Frame=8 Persons=1 (0 + 1) / 1]
Frame=10 Persons=1
Frame=11 Persons=1
Frame=9 Persons=1 (0 + 5) / 5]
[Stage 0:>

hdoop@lekhoi-VMware-Virtual-Platform: ~/D5200/Lab5
Sent frame_id=6
Sent frame_id=7
Sent frame_id=8
Sent frame_id=9
Sent frame_id=10
Sent frame_id=11
Sent frame_id=12
Sent frame_id=13
Sent frame_id=14
Sent frame_id=15
Sent frame_id=16
Sent frame_id=17
Sent frame_id=18
Sent frame_id=19
Sent frame_id=20
Sent frame_id=21
Sent frame_id=22
Sent frame_id=23
Sent frame_id=24
Sent frame_id=25
Sent frame_id=26
Sent frame_id=27
Sent frame_id=28
```

Output json :



The image shows a JSON output snippet in a terminal window. The JSON object contains frame_id, timestamp, person_count, and a list of bboxes. The first bbox is shown with its properties: x, y, w, h, and score.

```
{
  "frame_id": 1,
  "timestamp": "20260612_230934_514965",
  "person_count": 1,
  "bboxes": [
    {
      "x": 3,
      "y": 326,
      "w": 713,
      "h": 929,
      "score": 0.699
    }
  ]
}
```

Output khung hình tương ứng

